

planning will also help choose a unique and novel name for the app at this stage.

3. **UI/UX design:** This is the most important aspect to deliver a good and seamless user experience. The goal of UI/UX design is to create the best user experience while making the app interactive, easy to use and user-friendly. Under this, five aspects have to be covered.
 - (a) *Information architectures and workflows:* These determine the data the mobile app will display to the users, the data it will collect, user interactions and app information architecture.
 - (b) *Wireframes:* These are conceptual layouts to give visual structure to the app's functional requirements.
 - (c) *Style guide:* This determines the font, colour and brand that is to be reflected in the app design.
 - (d) *Mockups:* When the development of the app reaches the final stages, further modifications to its information architecture, workflow and aesthetics are required for the best user experience.
 - (e) *Prototype:* This is useful for the pre-launch testing of the app -- to find any bugs and to collect user feedback to make it better.
4. **App development:** In order to start the actual development, you need to define the technical architecture, technology to be used and the milestones for development. Any app development process covers three parts – the back-end, the API and the mobile front-end. The back-end includes the database and server-side objects that support the functions of the mobile app. The application programming interface (API) is the mode of communication between the app and the back-end server/database. The mobile front-end can be the user interface, which uses an API and a back-end to manage

the data. Various programming languages can be used for designing the front-end of the app.

5. **Testing:** Like in the case of testing any software for quality assurance, the mobile app development process too requires comprehensive quality testing to ensure the applications are stable, usable and secure. In order to do this, test cases are designed and the results are assessed. Software quality evaluation and tracking of fixes are done prior to retesting. Various types of testing can be performed — user experience testing, functional testing, performance testing, security testing, and device and platform testing.
6. **Deployment and support:** A mobile app has to be submitted to app stores like the Apple App store or Google Play for release. This requires a developer account as well as various metadata like the app title, a description, category, keywords, the launch icon, app store screenshots, etc. Once the app is submitted, it undergoes a review process that could range from a few days to several weeks, depending on the quality of the app and the guidelines-based procedure. And if the app is fine, it is made available and the developer can monitor its usage through mobile analytics platforms and key performance indicators (KPIs) to measure its success.

Overall, app development is a continuous process and requires regular feedback, feature updates and stability releases based on the user feedback.

Comparing Android and the iOS platform

Prior to developing an app, you need to decide on the platform to be chosen — Android or iOS. Table 2 highlights the differences between the two platforms. You can choose the platform depending on your requirements — whether it is for real-time client implementation or a freelancer based project.

Table 2

Basis of difference	Android	iOS
Programming language	Java	Swift/Objective-C
Source model and developer	Open source and managed by Google and the Open Handset Alliance	Developed and managed by Apple Inc.
IDE	Android Studio	XCODE
App launch and publishing time frame	A few hours after submission	A few days to a few weeks, depending on the review process prior to approval
Security	Collects and sells lots of user data to third party stores and is more vulnerable to attacks	Higher level of security with extra layers
Emulators	Slow emulators and the testing process is complex	Faster emulators and testing takes less time
Worldwide market share	Almost 75 per cent of the international market	Almost 23 per cent